

The Future of Stellar Populations with the Maunakea Spectroscopic Explorer (MSE)

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Abstract. The Maunakea Spectroscopic Explorer (MSE) is a massively multiplexed spectroscopic survey facility that is proposed to replace the Canada-France-Hawaii Telescope in the 2040s. Since 2019, due to the uncertainty for new facilities on Maunakea, the project has been focused on new technology enabling greater capabilities beyond the concept design reviewed facility plan. Enhanced fiber density, and thereby survey speed, is made possible by using a new quad mirror (QM) 11.5-meter telescope design with 18,000+ fibers and a 1.5 square degree field-of-view. The MSE spectrographs will be moderate-resolution (360 nm through H-band at $R=7,000$) and high-resolution ($R=40,000$). MSE's baseline NIR capabilities will enable studies of highly-reddened regions in the Local Group, unlike other proposed next generation facilities. The MSE large-scale survey instrument suite will enable the equivalent to a full SDSS Legacy Survey every several weeks. This work presents the current status of the project after the Fall 2024 MSE science Workshop.

Keywords. surveys, telescope, instrumentation: spectrographs, Galaxy: general, stars: general

1. Introduction

The Maunakea Spectroscopic Explorer (MSE; mse.cfht.hawaii.edu) project has eight science working groups, five of which are relevant for local group studies (Exoplanets and Stellar Astrophysics, Milky Way and Resolved Stellar Population, Chemical Nucleosynthesis, Astrophysical Tests of Dark Matter, and Time Domain Astronomy and Transients). This work presents the discussions related to these working groups and new technology and design work from the 2024 MSE Virtual Meeting held between October 28th and November 8th, 2024.

MSE's goal is to carry out the ultimate spectroscopic follow-up of the Gaia mission and is critical to our understanding of the faint and distant regimes of the Galaxy and Local Group. Uniquely, MSE will conduct in-situ chemodynamical analysis of individual stars in all Galactic components, searching for inter-relationships between them and for departures from equilibrium. MSE will bring about an entirely new era for nearby dwarf galaxy studies, enabling accurate chemo-dynamical measurements to be performed efficiently across the full range of dwarf galaxy luminosities ($10^{3-7} L_{\odot}$), and providing spectra for at least an order of magnitude more stars in each system. MSE will also provide a comprehensive understanding of the chemodynamics of M31 and M33.

This work comes after the workshop dedicated to updating the community on possible new technology designs, lessons learned from other large spectroscopic survey projects (SDSS, 4MOST, LAMOST, GALAH, DESI, PFS), and starting to evaluate should the existing MSE science requirements be updated based on new science since the 2019 science case (The MSE Science Team, 2019) and due to new science discoveries and the MSE project's delayed schedule.

毛纳基亚光谱探测器 (MSE) 与恒星种群的未来

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摘要. 毛纳基亚光谱探测器 (MSE) 是一个大规模多路复用光谱巡天设施, 计划在2040年代取代加拿大-法国-夏威夷望远镜。自2019年以来, 由于毛纳基亚新设施的不确定性, 该项目一直专注于开发新技术, 以实现超越概念设计审查设施计划的更大能力。通过采用新的四镜 (QM) 11.5米望远镜设计, 配备18,000+根光纤和1.5平方度的视场, 提高了光纤密度, 从而提升了巡天速度。MSE光谱仪将具备中等分辨率 (360纳米至H波段, $R=7,000$) 和高分辨率 ($R=40,000$)。MSE的基线近红外能力将使其能够研究本地群中高度红化的区域, 这是其他拟议的下一代设施所不具备的。MSE的大规模巡天仪器套件将能够在几周内完成相当于整个SDSS遗产巡天的任务。本文介绍了2024年秋季MSE科学研讨会后项目的最新进展。

关键词. {v*}veys, 望远镜, 仪器: 光谱仪, 星系: 一般, 恒星: 一般

1. 引言

毛纳基山光谱探测器 (MSE; mse.cfht.hawaii.edu) 项目设有八个科学工作组, 其中五个与本地星系群研究相关 (系外行星与恒星天体物理学、银河系与解析恒星群体、化学核合成、暗物质的天体物理测试、时域天文学与瞬变现象)。本文介绍了2024年10月28日至11月8日举行的MSE虚拟会议中与这些工作组相关的讨论, 以及新技术和设计工作的内容。

MSE的目标是执行盖亚任务的最终光谱后续观测, 对于我们理解银河系和本地群的微弱和遥远区域至关重要。独特的是, MSE将对所有银河系组成部分中的单个恒星进行原位化学动力学分析, 寻找它们之间的相互关系以及偏离平衡的情况。MSE将为附近矮星系的研究带来一个全新的时代, 使得在整个矮星系光度范围内 ($10^{-7} L_{\odot}$) 能够高效地进行精确的化学动力学测量, 并为每个系统中的恒星提供至少一个数量级的光谱。MSE还将全面理解M31和M33的化学动力学。

这项工作是在一次专门向社区更新可能的新技术设计、从其他大型光谱巡天项目 (SDSS、4MOST、LAMOST、GALAH、DESI、PFS) 中吸取的经验教训的研讨会之后进行的, 并开始评估是否应根据自2019年科学案例 (MSE科学团队, 2019) 以来的新科学发现以及MSE项目延迟的时间表, 更新现有的MSE科学要求。

2. MSE New Design work

MSE has completed the concept design review, however due to delays with Maunakea, we have reopened consideration of the design of the telescope, focal plane, and spectrographs, which has led to the Quad-Mirror design (that enables a five fold increase in fiber density; Barden et al 2024; Figure 1 *left*). This new design leads to a number of key improvements, (1) allows for a larger primary, primarily limited due to size of the largest transmissive optical element, (2) move the fiber focal plane from prime-focus to Nasmyth focus enabling a physically larger focal plane, (3) the Nasmyth focal plane enables packing 18,000+ fibers, and (4) the Nasmyth focus allows the possibility of significantly shorter fibers for spectrographs that could be placed on the nasmyth platform, which could allow ~ 1 magnitude improvement in near-UV throughput.

Due to difficulty in producing the needed high-resolution spectrographs and to take advantage to the potential of increased blue throughput, MSE work has been done to develop and test the concept of pupil and wavelength splitting (Barden 2024; Figure 1 *right*). Through a NSF ATI grant (NSF-2307501), Sam Barden has developed and tested a prototype pupil and wavelength splitting system. The potential of these systems are two-fold, (1) wavelength splitting at the fiber level enables (2) pupil splitting enables the each fiber to "see" a smaller pupil enabling the use of smaller and cheaper spectrographs, but also requiring more of them. Pupil and wavelength splitting can be implemented separately or on combination, and we expect MSE and other future projects to consider implementing these technologies.

The current baseline MSE QM-based instrument/focal plane configuration includes $\sim 14,000+$ fibers feeding 5-6 channel medium-resolution (MR) spectrographs covering the full $0.4\text{--}1.3\mu\text{m}$ and H -band ($1.5\text{--}1.7\mu\text{m}$) with $R \sim 5000$ and $4000+$ fibers feeding high-resolution (HR) spectrographs with 3 channels ($401\text{--}471\text{ nm}$ @ $R = 40k$, $471\text{--}489\text{ nm}$ @ $R = 40k$, and $625\text{--}674\text{ nm}$ @ $R = 20k$). At the workshop, new possible MSE MR and LR spectrograph designs were presented, including a folded Schmidt camera design, that would enable less optical obstruction and smaller/easier detector mounting (Will Saunders, private communication).

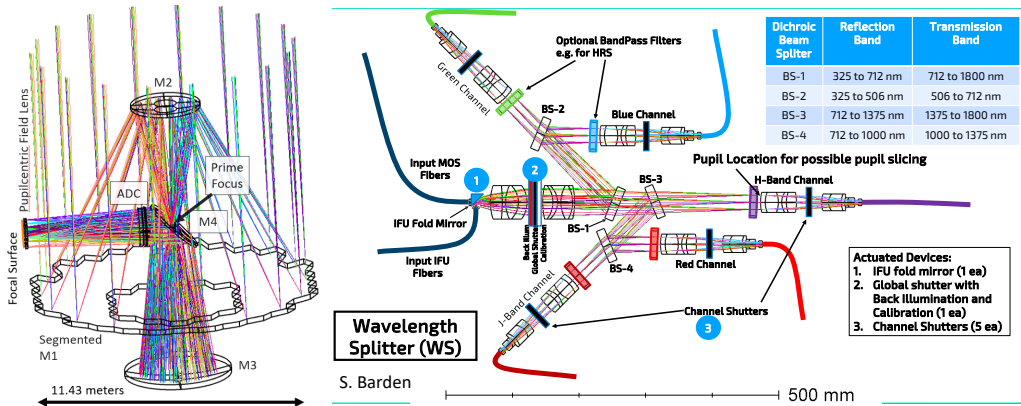


Figure 1. (Left) The MSE Quad Mirror (QM) design that is the now current proposed baseline for the MSE telescope, designed by Sam Barden (Barden 2024). This design could be scaled up, but is currently constrained considering working within the existing CFHT footprint and by realistic assessment of the producibility of largest optical element. (Right) The new proposed pupil and wavelength-splitting technology for MSE, which will enable smaller and cheaper, but more, spectrographs to be built to meet the MSE science requirements. This viability of this technology is being evaluated by Sam Barden at CFHT supported by an NSF ATI grant (AST-2307501; Barden et al. 2024).

2. MSE新设计工作

MSE已完成概念设计审查，但由于莫纳克亚山的延误，我们重新考虑了望远镜、焦平面和光谱仪的设计，这导致了四镜设计（使光纤密度增加五倍；Barden等，2024年；图1 left）。这一新设计带来了多项关键改进：（1）允许更大的主镜，主要受限于最大透射光学元件的尺寸；（2）将光纤焦平面从主焦点移至Nasmyth焦点，从而实现物理上更大的焦平面；（3）Nasmyth焦平面可容纳18,000+根光纤；（4）Nasmyth焦点允许光谱仪使用显著更短的光纤，这些光纤可以放置在Nasmyth平台上，这可能使近紫外通量提高~1个星等。

由于制造所需的高分辨率光谱仪存在困难，并且为了利用增加蓝色通量的潜力，MSE工作已经开展，以开发和测试光瞳和波长分裂的概念（Barden 2024；图1 right）。通过NSF ATI资助（NSF-2307501），Sam Barden开发并测试了一个光瞳和波长分裂系统的原型。这些系统的潜力是双重的，（1）在光纤级别进行波长分裂使得（2）光瞳分裂使每根光纤“看到”一个更小的光瞳，从而能够使用更小、更便宜的光谱仪，但也需要更多的光谱仪。光瞳和波长分裂可以单独或组合实施，我们预计MSE和其他未来项目将考虑采用这些技术。

当前基于MSE QM的基线仪器/焦平面配置包括~ 14,000+根光纤，用于5-6通道的中分辨率（MR）光谱仪，覆盖全波段0.4-1.3 μm 和H波段（1.5-1.7 μm ），以及R~5000和4000+根光纤，用于高分辨率（HR）光谱仪，具有3个通道（401-471 nm @ R = 40k, 471-489 nm @ R = 40k, 和625-674 nm @ R = 20k）。在研讨会上，展示了新的MSE MR和LR光谱仪设计方案，包括一种折叠式施密特相机设计，该设计可以减少光学遮挡并简化探测器的安装（Will Saunders，私人通信）。

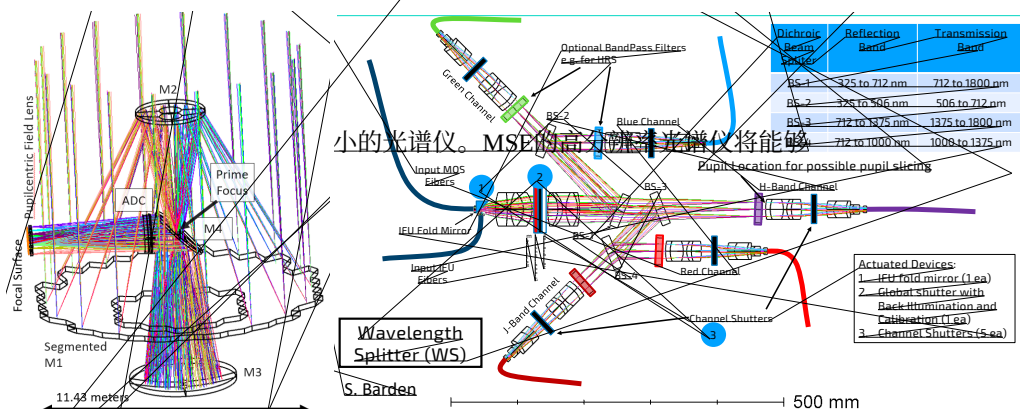


图1. (左) MSE四镜 (QM) 设计，这是目前为MSE望远镜提出的基线设计，由Sam Barden设计（Barden 2024）。该设计可以按比例放大，但目前考虑到在现有CFHT占地面积内工作以及对最大光学元件可生产性的现实评估，设计受到限制。(右) 为MSE提出的新光瞳和波长分离技术，该技术将使得能够建造更小、更便宜但更多的光谱仪，以满足MSE的科学要求。该技术的可行性正在由Sam Barden在CFHT进行评估，并得到NSF ATI资助（AST-2307501；Barden等，2024）的支持。

New potential fiber positioner designs, with significantly larger positioner patrol range were presented (Roloef de Jong, private communication) that are currently starting prototype development. This or similar technology could allow MSE to better populate the high resolution (HR) fibers, while also reduce the total number of HR spectrographs needed per wavelength channel.

3. MSE Workshop 2024 - Stellar Populations Science Implications

The key stellar populations science questions discussed and their implications on MSE science requirement/design work:

- What is the optimal High-Resolution (HR) fiber density, considering potential new positioner technologies, target densities, and the physically larger QM focal plane.
- Evaluate trade-off between the number of HR fibers in the focal plane (currently $\sim 1/3$ of the fibers would be HR to allow coverage over the full focal plane) and how many HR spectrograph channels/wavelength regions (currently only 3 HR channels) should be covered to produce the groundbreaking science in the 2040s.
- Determine what HR resolution is *needed* ($R \sim 20,000\text{--}80,000$) to reliably measure key enrichment tracing abundances (examples included Fe, Zn, Mn, U, Th, C, N) to the desired precision. Particularly taking into account that at different metallicities will require different resolutions, e.g., line blanketing in metal-rich stars ($[\text{Fe}/\text{H}] \geq 0$) may actually require higher resolution than metal-poor ($[\text{Fe}/\text{H}] < 2$) stars.
- For the medium resolution (MR) spectrographs is $R \sim 5000$ sufficient for stellar population science needs? Would one channel need to be higher resolution to enable more science, e.g., higher precision velocities or more elements ($[\alpha/\text{Fe}]$, can other elements be measured, carbon, nitrogen, oxygen from molecules with NIR at $R \sim 5000$)?
- What new science is enabled by taking advantage of the Maunakea site through using the Near-UV (310-390 nm) sky?
- Is the science enabled by adding high resolution Near-UV and/or *H*-band science worth the additional costs (UV optimized spectrograph, cryogenic NIR spectrograph), and potentially increased detector cost? Does wavelength splitting work well enough to enable these options?

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新的潜在光纤定位器设计，具有显著更大的定位器巡逻范围，已经提出（Roloef de Jong, 私人通信），目前正在开始原型开发。这种或类似的技术可以使MSE更好地填充高分辨率（HR）光纤，同时减少每个波长通道所需的HR光谱仪总数。

3. MSE研讨会2024 - 恒星种群科学影响

讨论的关键恒星种群科学问题及其对MSE科学需求/设计工作的影响：

- 考虑到潜在的新型定位器技术、目标密度以及物理上更大的QM焦平面，最佳高分辨率（HR）光纤密度{ ν^* }是多少。

- 评估焦平面中高分辨率（HR）光纤数量（目前 $\sim 1/3$ 的光纤为HR，以实现对整个焦平面的覆盖）与应覆盖多少HR光谱仪通道/波长区域（目前仅有3个HR通道）之间的权衡，以在2040年代产生突破性的科学成果。

- 确定HR分辨率needed ($R \sim 20,000-80,000$)，以可靠地测量关键富集追踪丰度（包括Fe、Zn、Mn、U、Th、C、N等示例）至所需精度。特别考虑到不同金属丰度将需要不同的分辨率，例如，金属丰度高的恒星（ $[\text{Fe}/\text{H}] \geq 0$ ）可能实际上比金属丰度低的恒星（ $[\text{Fe}/\text{H}] < 2$ ）需要更高的分辨率。

- 对于中等分辨率（MR）光谱仪， $R \sim 5000$ 是否足以满足恒星群体科学的需求？是否需要一个通道具有更高的分辨率以实现更多的科学研究，例如更高精度的速度测量或更多元素的测量（ $[\alpha/\text{Fe}]$ ，是否可以在近红外波段 $R \sim 5000$ 处测量其他元素，如碳、氮、氧等分子）？

- 利用毛纳基山站点，通过使用近紫外（310-390 nm）天空，能够开启哪些新的科学研究？

- 增加高分辨率近紫外和/或H波段科学所带来的科学价值是否值得额外的成本（紫外优化光谱仪、低温近红外光谱仪）以及可能增加的探测器成本？波长分光技术是否足够有效以支持这些选项？

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